

Elliptic curves over finite fields

Ex 9.1

$$E = \{(x, y) \in \mathbb{Z}_p^2 \mid y^2 = x^3 + ax + b\} \cup \{0\}$$

$$p=11, a=3, b=1$$

$$E = \{(x, y) \in \mathbb{Z}_{11}^2 : y^2 = x^3 + 3x + 1\} \cup \{0\}$$

$$\Delta(E) = -16(4 \cdot 3^3 + 27 \cdot 1) = -2160 \not\equiv 0 \pmod{11}$$

Non-singular curve

$$E(\mathbb{Z}_{11}) = ?$$

x	0	1	2	3	4	5	6	7	8	9	10
y^2	1	5	15	37	77	141	235	365	537	757	10003
$y^2 \pmod{11}$	1	5	4	4	0	9	4	2	9	9	10
y	1, 10	4, 7	2, 9	2, 9	0	3, 8	2, 9	-	3, 8	3, 8	-

$$E(\mathbb{Z}_{11}) = \{(0, 1), (0, 10), (1, 4), (1, 7), (2, 2), (2, 9), (3, 2), (3, 9), (4, 0), (6, 2), (6, 9), (8, 3), (8, 8), (9, 3), (9, 8), (5, 3), (5, 8)\} \cup \{0\}$$

Ex 9.2 $p=11, E_2(\mathbb{Z}_{11}) = \{(x, y) \in \mathbb{Z}_{11}^2 \mid y^2 = x^3 + 4x + 5\} \cup \{0\}$

Verify: $\Delta(E) = 7 \pmod{11} (\neq 0)$ non-singular,

$$|E_2(\mathbb{Z}_{11})| = 8$$

$$GL(n, \mathbb{C}) = \{A \in M_n(\mathbb{C}) \mid A \text{ is invertible}\}$$

$$x^n + y^n = z^n, \quad n \geq 3$$

E.g. $p=2$ $E(\mathbb{Z}_2) = \{ (x,y) \in \mathbb{Z}_2^2 \mid y^2+y = \underline{x^3+x} \} \cup \{\infty\}$

x	$y^2+y=?$	y
0	0	0,1
1	0	0,1

$\therefore \{ (0,0), (0,1), (1,0), (1,1) \} \cup \{\infty\}$

$|E(\mathbb{Z}_2)| = \underline{5}$

E.g. $p=2$, $E(\mathbb{Z}_2) = \{ (x,y) \in \mathbb{Z}_2^2 \mid y^2+y = x^3+1 \} \cup \{\infty\}$

x	$y^2+y=x^3+1$	y
0	$y^2=1$	1
1	$y^2+y=0$	0,1

$E(\mathbb{Z}_2) = \{ (0,1), (1,0), (1,1) \} \cup \{\infty\}$, $\curvearrowright$ 4 pts.

Char $\mathbb{F} = 2$ but $\mathbb{F} \neq \mathbb{Z}_2$

$\mathbb{F}_4 = \{ 0,1,\alpha,\alpha^2 \}$ $\alpha^2 = 1+\alpha$

$E(\mathbb{F}_4) = \{ (x,y) \in \mathbb{F}_4^2 \mid y^2+y = x^3+x \} \cup \{\infty\}$

x	$y^2+y=x^3+x$	y
0	0	0,1
1	0	0,1
α	α^2	—
$\alpha^2=1+\alpha$	α	—

$$\begin{aligned} y^2 + y &= \alpha^3 + \alpha = \alpha(1 + \alpha) + \alpha \\ &= \alpha + \alpha^2 + \alpha \\ &= 1 + \alpha = \alpha^2 \end{aligned}$$

$$(a+b)^2 = a^2 + b^2$$

$$(1+\alpha)^2 = (\alpha^2)^2 = \alpha$$

$$\begin{aligned} \alpha^3 &= 1 \\ \alpha(\alpha^2) &= \alpha(1 + \alpha) \\ &= \alpha + \alpha^2 \\ &= \alpha + 1 + \alpha \\ &= 1 \end{aligned}$$

$$\begin{aligned} y^2 + y &= (\alpha^2)^3 + \alpha^2 = (1 + \alpha)^3 + \alpha^2 \\ &= (\alpha^2)^2 \alpha^2 + \alpha^2 \\ &= \alpha \alpha^2 + \alpha^2 \\ &= \alpha^3 + \alpha^2 + 1 \\ &= 1 + \alpha + 1 \\ &= \alpha \end{aligned}$$

$$y^2 + y = 0, \alpha, \alpha^2$$

$$0^2 + 0 = 0, 1^2 + 1 = 0$$

$$\alpha^2 + \alpha = 1 + \alpha + \alpha = 1$$

$$(\alpha^2)^2 + \alpha^2 = \alpha + \alpha^2 = \alpha + 1 + \alpha = 1$$

$$\{(0,0), (0,1), (1,0), (1,1), \infty\} = E(\mathbb{F}_4)$$

$$\text{E.g., } E(\mathbb{F}_4) = \{(x,y) \in \mathbb{F}_4^2 \mid y^2 + xy = x^3 + 1\} \cup \{\infty\}$$

x	$y^2 + xy = x^3 + 1$	y
0	$y^2 = 1$	1
1	$y^2 + y = 0$	0, 1
α	$y^2 + \alpha y = 0$	0, α
α^2	$y^2 + \alpha^2 y = 0$	0, α^2

$$E(\mathbb{F}_4) = \{(0,1), (1,0), (1,1), (\alpha,0), (\alpha,\alpha), (\alpha^2,0), (\alpha^2,\alpha^2), \infty\}$$

P-3 Chen 3

$$E_1(\mathbb{F}_3) = \{(x,y) \in \mathbb{F}_3^2 \mid y^2 = x^3 - x + 2\} \cup \{\infty\}$$

is singular.

$$E_2(\mathbb{Z}_3) = \{ (x, y) \in \mathbb{Z}_3^2 \mid y^2 = x^3 - x + 2 \} \cup \{\infty\}$$

$$= \{ \emptyset, \infty \} \quad \text{two pts only}$$

$$E_3(\mathbb{F}_9) = \{ (x, y) \in \mathbb{F}_9 : y^2 = x^3 - x + 2 \} \cup \{\infty\}$$

$$\mathbb{F}_9 = \{ \alpha^i : 0 \leq i \leq 8 \} \quad \underline{\alpha^9 = 1}$$

$$\alpha \text{ satisfies } \underline{\alpha^2 + \alpha + 2 = 0}$$

Find all pts of the

Elliptic curve.

$$\mathbb{F}_9 = \frac{\mathbb{Z}_3[x]}{\langle x^2 + x + 2 \rangle}$$

7 pts